

Advanced Topics in Computer Graphics II

Geometry Processing

Mesh Parameterization

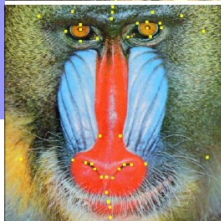
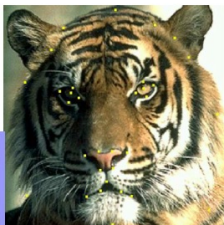
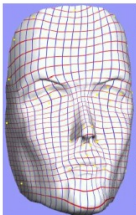


December 10, 2024



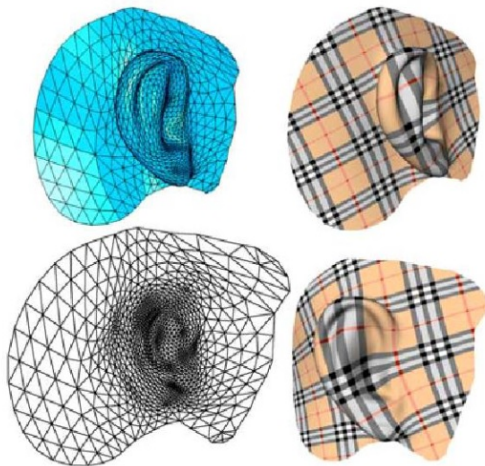


Problem: Map a texture to a mesh





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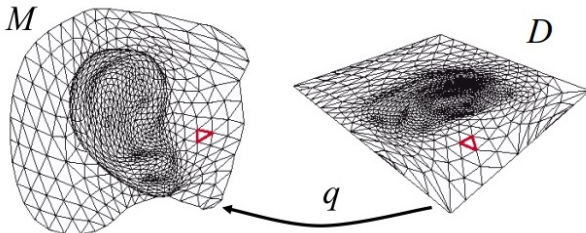
Given: A triangle mesh $\mathcal{M}$ consisting of faces $\mathcal{F}$, vertices $\mathcal{V}$ and edges $\mathcal{E}$:

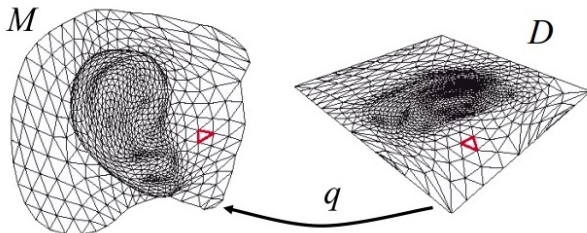
$$\mathcal{M} = (\mathcal{F}, \mathcal{V}, \mathcal{E})$$

$$\mathcal{F} = \{\mathbf{f}_i = (\mathbf{v}_{i1}, \mathbf{v}_{i2}, \mathbf{v}_{i3}) \in \mathbb{N}^3\} \quad , \quad \mathcal{V} = \{\mathbf{v}_i \in \mathbb{R}^3\} \quad , \quad \mathcal{E} = \{(\mathbf{v}_{i1}, \mathbf{v}_{i2}) \in \mathbb{N}^2\}$$

Task: Find a parameterization

$$\mathbf{q}: \mathbb{R}^2 \supset \mathcal{D} \rightarrow \mathbb{R}^3$$





Idea: Triangulated surfaces (meshes) are naturally parametrized by piece-wise linear functions. Their pieces correspond to the triangles of the mesh.

→ If we know the mapping q at the vertices, we also know it inside the triangle using barycentric coordinates $(\lambda_i, \lambda_j, \lambda_k)$:

$$q(u, v) = \lambda_i \mathbf{v}_i + \lambda_j \mathbf{v}_j + \lambda_k \mathbf{v}_k$$

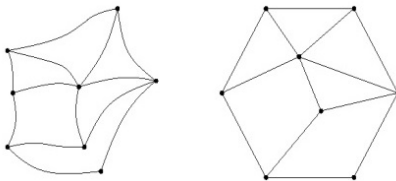
where (i, j, k) is the triplet such that the triangle $[(u_i, v_i), (u_j, v_j), (u_k, v_k)]$ in parameter space contains the point (u, v) .



Task: Find a parameterization q of the vertices:

$$q(u_i, v_i) = v_i$$

Idea: Imagine how a graph could be drawn¹



Theorem (Parameterization)

Let $\mathcal{M}$ be a triangulated surface homeomorphic to a disk. If the coordinates (u, v) of the boundary vertices lie on a convex polygon, and if the coordinates of the internal nodes are convex combinations of their neighbors, then the parameterization formed by (u, v) is valid.

¹W. T. Tutte. "How to draw a graph". In: *Proc Lond Math Soc* 13 (1963), pp. 743–767.



According to the mentioned theorem, we can compute a parameterization using the following steps:

1. Separate the set of vertices $\mathcal{V}$ into boundary and interior vertices

$$\mathcal{V} = \underbrace{\{\mathbf{v}_1, \dots, \mathbf{v}_n\}}_{\text{interior}} \cup \underbrace{\{\mathbf{v}_{n+1}, \dots, \mathbf{v}_N\}}_{\text{boundary}}$$

2. Choose parameter values (u_i, v_i) , $i = n+1, \dots, N$ for the boundary vertices $\{\mathbf{v}_{n+1}, \dots, \mathbf{v}_N\}$
3. For each $i \in \{1, \dots, n\}$ choose any set of real numbers λ_{ij} for $j = 1, \dots, N$ such that

$$\lambda_{ij} = 0 \text{ for } (i, j) \notin E \quad , \quad \lambda_{ij} \geq 0 \text{ for } (i, j) \in E \quad , \quad \sum_{j=1}^N \lambda_{ij} = 1$$

4. Compute parameter values (u_i, v_i) , $i = 1, \dots, n$ for the interior vertices by solving the linear equations system of convex combinations

$$(u_i, v_i) = \sum_{j=1}^N \lambda_{ij} \cdot (u_j, v_j) \quad , \quad i = 1, \dots, n$$



Problem: How to solve the equation system?

$$(u_i, v_i) = \sum_{j=1}^N \lambda_{ij} \cdot (u_j, v_j) \quad , \quad i = 1, \dots, n$$

Solution: Split the sum into (unknown) interior and (known) boundary terms:

$$\underbrace{(u_i, v_i)}_{\text{unknown}} - \sum_{j=1}^n \lambda_{ij} \cdot \underbrace{(u_j, v_j)}_{\text{unknown}} = \sum_{j=n+1}^N \lambda_{ij} \cdot \underbrace{(u_j, v_j)}_{\text{known}}$$

In matrix form we get

$$\begin{pmatrix} 1 & -\lambda_{12} & \cdots & -\lambda_{1n} \\ & \ddots & & \\ & & \ddots & \\ -\lambda_{n1} & \cdots & -\lambda_{n(n-1)} & 1 \end{pmatrix} \begin{pmatrix} u_1 & v_1 \\ \vdots & \vdots \\ \vdots & \vdots \\ u_n & v_n \end{pmatrix} = \begin{pmatrix} \lambda_{1(n+1)} & \cdots & \lambda_{1N} \\ & \ddots & \\ \lambda_{n(n+1)} & \cdots & \lambda_{nN} \end{pmatrix} \begin{pmatrix} u_{n+1} & v_{n+1} \\ \vdots & \vdots \\ u_N & v_N \end{pmatrix}$$

Note: The left matrix is a Laplace Matrix with negative signs.



This system can be solved separately for u and v :²

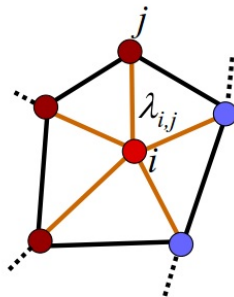
$$\mathbf{A} \mathbf{u} = \mathbf{u}_b \quad , \quad \mathbf{A} \mathbf{v} = \mathbf{v}_b$$

→ Since $\mathbf{A}$ is a Laplace Matrix, it can be efficiently constructed in parallel (see Mesh Smoothing lecture).

Theorem (Floater)

Given a triangulated surface. If all weights $\lambda_{ij} > 0$, then the matrix $\mathbf{A}$ is non-singular, diagonally dominant and sparse.

Therefore, the linear equations system as defined above has a unique solution. This solution is a legal parameterization of the triangulated surface.



²M. S. Floater. "Parametrization and smooth approximation of surface triangulations". In: *Computer aided geometric design* 14.3 (1997), pp. 231–250.



Question: How to choose the weights λ_{ij} ?

Idea: Define coordinate weights and use: $\lambda_{ij} = \frac{w_{ij}}{\sum_j w_{ij}}$

- Uniform coordinates (Tutte, 1963)

$$w_{ij} = 1$$

- Wachspress coordinates (Wachspress, 1975)

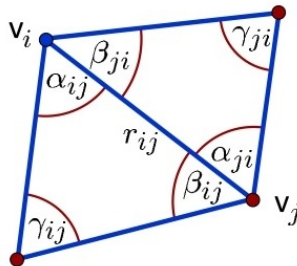
$$w_j = \frac{\cot \alpha_{ji} + \cot \beta_{ij}}{r_{ij}^2}$$

- Discrete Harmonic coordinates (Iserles, 1996)

$$w_{ij} = \cot \gamma_{ij} + \cot \gamma_{ji}$$

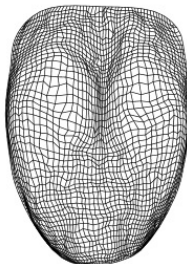
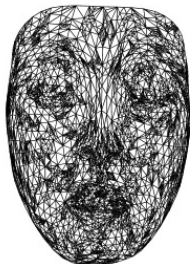
- Mean Value coordinates (Floater, 2003)

$$w_{ij} = \frac{\tan\left(\frac{\alpha_{ij}}{2}\right) + \tan\left(\frac{\beta_{ji}}{2}\right)}{r_{ij}}$$

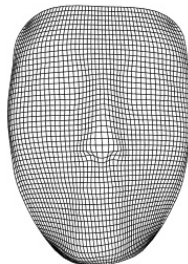




Results:



Uniform



Mean Value

Limitations:

- ▶ Boundary is fixed manually at the beginning. Better choices?
- ▶ Some weights (defined above) are not positive
→ parameterization might be not legal!



Idea: Define the parameterization in a way that conformality is ensured!

Consider a small disk $\mathcal{C}$ at point $p_0 = (u, v)$:

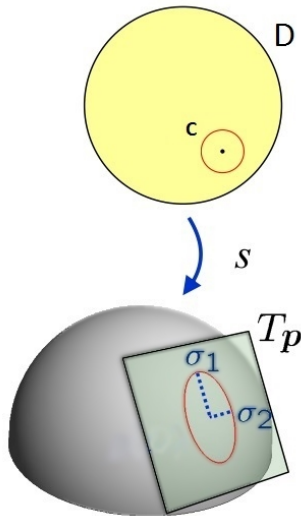
$$\mathcal{C}(p_0) = \{p \in \mathcal{D} \mid \|p - p_0\| < \epsilon\}$$

A vector w in the disk is stretched on the surface by the Jacobian (Differential Geometry):

$$w_{TP} = J w \quad , \quad J \in \mathbb{R}^{3 \times 2}$$

We can decompose it to

$$J = U \Sigma V^T$$

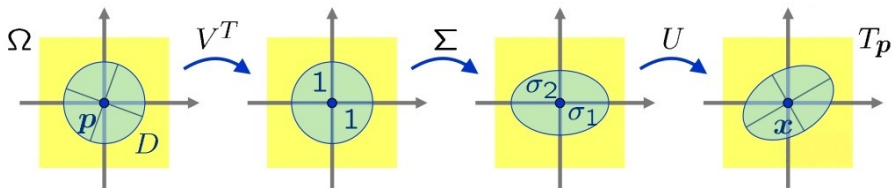




We have

$$\mathbf{J} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T = \mathbf{U} \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{pmatrix} \mathbf{V}^T$$

$$\mathbf{U} \in \mathbb{R}^{3 \times 3} \quad , \quad \sigma_1 \geq \sigma_2 > 0 \quad , \quad \mathbf{V} \in \mathbb{R}^{2 \times 2}$$



So we can measure distortion/conformality by computing the singular values of $\mathbf{J}$. Since these are the square-roots of the eigenvalues of $\mathbf{M}_1 = \mathbf{J}^T \mathbf{J} = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, we can calculate them using the formula

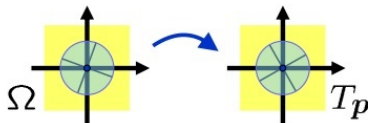
$$\sigma_{1,2} = \sqrt{\lambda_{1,2}} \quad , \quad \lambda_{1,2} = \frac{1}{2} \left((E + G) \pm \sqrt{4F^2 + (E - G)^2} \right)$$



Classifications:

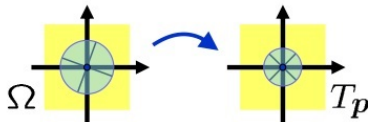
- Isometric or length-preserving

$$\sigma_1 = \sigma_2 = 1$$



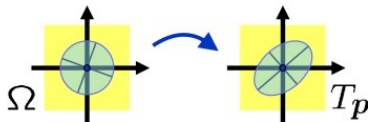
- Conformal or angle-preserving

$$\sigma_1 = \sigma_2$$



- Equiareal or area-preserving

$$\sigma_1 \cdot \sigma_2 = 1$$



Note: Isometric $\Leftrightarrow$ Conformal + Equiareal



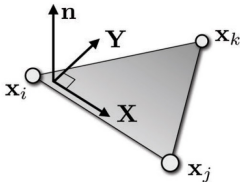
Idea: Define conformal maps on the mesh and minimize an energy.

- Conformality means that the tangent vectors $\mathbf{q}_u$ and $\mathbf{q}_v$ are orthogonal and have the same length. This can be written as

$$\mathbf{q}_v = \mathbf{n} \times \mathbf{q}_u$$

- If a function is conformal, this also holds for its inverse (Jacobian of the inverse function is the inverse Jacobian of the original function)

Now consider the gradient in a triangle:

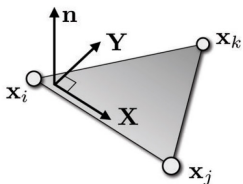


$$\mathbf{X} = \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|}$$

$$\mathbf{n} = \frac{\mathbf{X} \times (\mathbf{x}_k - \mathbf{x}_i)}{\|\mathbf{X} \times (\mathbf{x}_k - \mathbf{x}_i)\|}$$

$$\mathbf{Y} = \mathbf{n} \times \mathbf{X}$$

Now consider the gradient in a triangle:



$$\mathbf{X} = \frac{\mathbf{x}_j - \mathbf{x}_i}{\|\mathbf{x}_j - \mathbf{x}_i\|}$$

$$\mathbf{n} = \frac{\mathbf{X} \times (\mathbf{x}_k - \mathbf{x}_i)}{\|\mathbf{X} \times (\mathbf{x}_k - \mathbf{x}_i)\|}$$

$$\mathbf{Y} = \mathbf{n} \times \mathbf{X}$$

By studying the inverse of the parameterization, we see that

$$\nabla \mathbf{u} = \begin{pmatrix} \frac{\partial \mathbf{u}}{\partial X} \\ \frac{\partial \mathbf{u}}{\partial Y} \end{pmatrix} = \frac{1}{2 A_T} \underbrace{\begin{pmatrix} Y_j - Y_k & Y_k - Y_i & Y_i - Y_j \\ X_k - X_j & X_i - X_k & X_j - X_i \end{pmatrix}}_{=: M_T} \begin{pmatrix} u_i \\ u_j \\ u_k \end{pmatrix}$$



Since the inverse of a conformal map is also a conformal map, we can write the conformality condition as

$$\nabla \mathbf{v} = \mathbf{n} \times \nabla \mathbf{u}$$

Another more practical notation is

$$\nabla \mathbf{v} = (\nabla \mathbf{u})^\perp = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \nabla \mathbf{u}$$

If we now plug the gradient of a triangle in this formula, we obtain

$$\mathbf{M}_T \begin{pmatrix} v_i \\ v_j \\ v_k \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{M}_T \begin{pmatrix} u_i \\ u_j \\ u_k \end{pmatrix}$$

Therefore,

$$\left\| \mathbf{M}_T \begin{pmatrix} v_i \\ v_j \\ v_k \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{M}_T \begin{pmatrix} u_i \\ u_j \\ u_k \end{pmatrix} \right\|^2$$

measures the deviation of the parametrization of the triangle $(\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k)$ from a conformal one.



To find the best possible conformal parametrization of the given mesh we minimize the quadratic 'Least Squares Conformal Energy':

$$E_{LSCM} = \sum_{T=(i,j,k)} \left\| M_T \begin{pmatrix} v_i \\ v_j \\ v_k \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} M_T \begin{pmatrix} u_i \\ u_j \\ u_k \end{pmatrix} \right\|^2$$

Note: E_{LSCM} is invariant against arbitrary translations and rotations applied to the parameter space. To resolve this, at least 2 vertices have to be fixed!



Theorem (Riemann³)

Any continuous surface has a conformal parameterization.

Theorem (Least Squares Conformal Maps⁴)

Given a triangulated surface. If the Gaussian curvature vanishes at every point, then there exists a conformal parameterization. The surface is then called developable. Otherwise the minimized error

$$E_{LSCM} = \sum_{T=(i,j,k)} \left\| M_T \begin{pmatrix} v_i \\ v_j \\ v_k \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} M_T \begin{pmatrix} u_i \\ u_j \\ u_k \end{pmatrix} \right\|^2$$

is not zero and LSCM computes the best parameterization with minimal distortion. The surface is then called non-developable.

³M. Berger. *A panoramic view of Riemannian geometry*. Berlin, New York, Heidelberg: Springer, 2003.

⁴B. Lévy et al. "Least squares conformal maps for automatic texture atlas generation". In: *ACM Transactions on Graphics (TOG)*. vol. 21. 3. ACM. 2002, pp. 362–371.



Example:

Developable



Zero

Gaussian Curvature

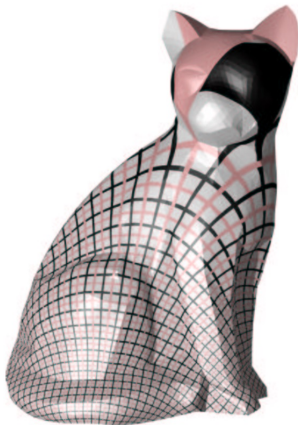
Non-Developable



Non-Zero

Gaussian Curvature

Results:



Limitation: Area stretched quite large at some regions!

Observation: LSCM produces parameterizations that preserve angles as good as possible at the cost of large area distortions!

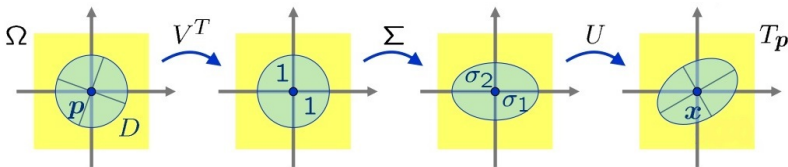
Idea: Try to minimize both:

$$E = \int_S e(v) dS$$

where S is a surface (mesh) and $e(v)$ the local energy at point $v \in S$.

Remember: Distortion can be measured using the Jacobian

$$J = U \Sigma V^T = U \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{pmatrix} V^T$$



Classification:

- Isometric or length-preserving

$$\sigma_1 = \sigma_2 = 1$$

- Conformal or angle-preserving

$$\sigma_1 = \sigma_2 \Leftrightarrow \frac{\sigma_1}{\sigma_2} = 1$$

- Equiareal or area-preserving

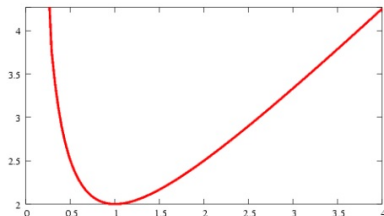
$$\sigma_1 \cdot \sigma_2 = 1$$

Note: Isometric $\Leftrightarrow$ Conformal + Equiareal

For the local energy term e , use the following function:

$$f(x) = x + \frac{1}{x}$$

This function is convex, has a minimum at $x = 1$ and goes to infinity for $x \rightarrow 0$ and $x \rightarrow \infty$.



So use this function f to define an energy for area and angle distortion:

- Area energy

$$e_{area} = \sigma_1 \cdot \sigma_2 + \frac{1}{\sigma_1 \cdot \sigma_2}$$

- Angle energy

$$e_{angle} = \frac{\sigma_1}{\sigma_2} + \frac{\sigma_2}{\sigma_1}$$

→ This is precisely the **MIPS** (**M**ost **I**sometric **P**arameterization**S**) energy defined by Hormann and Greiner (2000)⁵.

Now, we can define the total energy as

$$e_{combined} = e_{angle}^{1-\theta} \cdot e_{area}^{\theta}$$

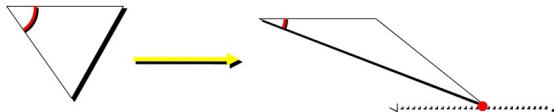
where θ is a user-controlled parameter describing the importance of each term.

⁵K. Hormann and G. Greiner. *MIPS: An efficient global parametrization method*. Tech. rep. DTIC Document, 2000.

The total energy is defined as

$$e_{combined} = e_{angle}^{1-\theta} \cdot e_{area}^{\theta}$$

Problem: Area is invariant against shear transformations $\rightarrow$ numerical problems



Therefore drop the exponent $1 - \theta$ leading to the final energy function⁶:

$$e_{combined} = e_{angle} \cdot e_{area}^{\theta}$$

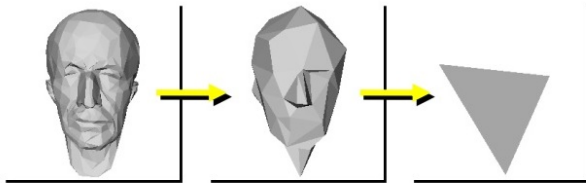
which leads to the following interpretation:

- ▶ $\theta = 0$: Angle preservation $\rightarrow$ MIPS, Conformal Maps , ...
- ▶ $\theta = 1$: Length preservation $\rightarrow e_{combined} = e_{angle} \cdot e_{area} = f(\sigma_1^2) \cdot f(\sigma_2^2)$
- ▶ $\theta \rightarrow \infty$: Area preservation

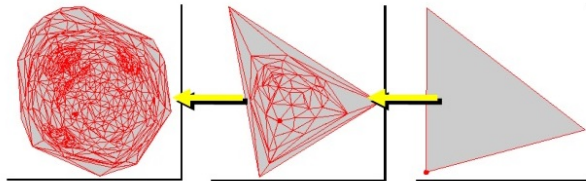
⁶P. Degener et al. "An Adaptable Surface Parameterization Method.". In: *IMR 3* (2003), pp. 201–213.

The total energy is minimized using a hierarchical approach⁷:

1. Build sequence of Split Sets

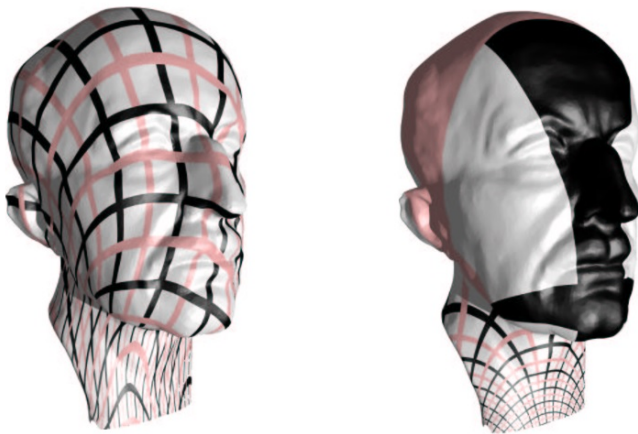


2. Relax



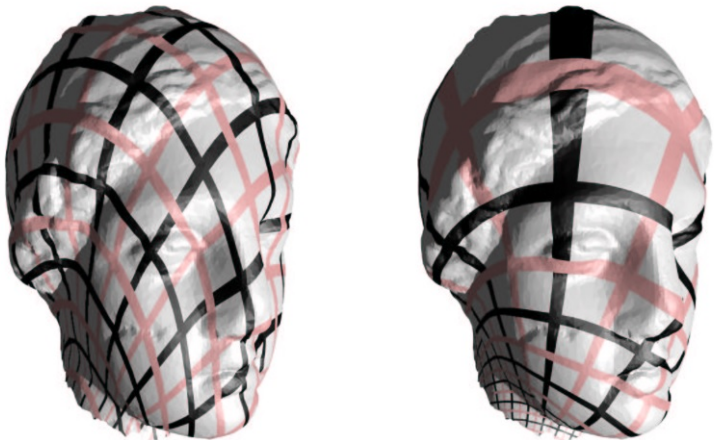
⁷K. Hormann, G. Greiner, and S. Campagna. "Hierarchical parametrization of triangulated surfaces". In: *Vision, Modeling, and Visualization*. Vol. 1999. 1999, p. 219.

Results:



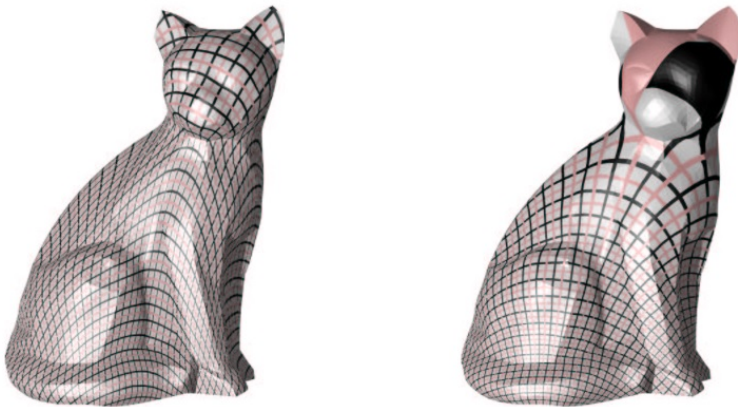
Left : ASF with $\theta = 1.0$, **Right** : LSCM

Results:



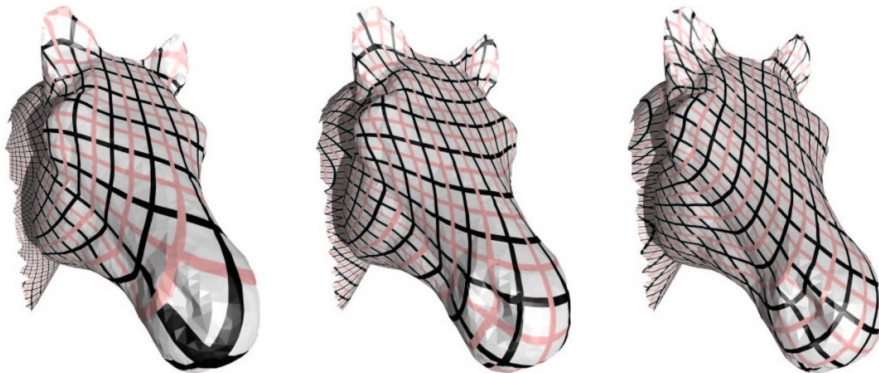
Left : ASF with $\theta = 1.0$, **Right :** LSCM

Results:



Left : ASF with $\theta = 1.0$, **Right** : LSCM

Results:



Left : $\theta = 0.3$, **Middle** : $\theta = 1.0$, **Right** : $\theta = 3.0$



- [1] M. Berger. *A panoramic view of Riemannian geometry*. Berlin, New York, Heidelberg: Springer, 2003.
- [2] P. Degener, J. Meseth, and R. Klein. "An Adaptable Surface Parameterization Method.". In: *IMR 3* (2003), pp. 201–213.
- [3] M. S. Floater. "Mean value coordinates". In: *Computer aided geometric design* 20.1 (2003), pp. 19–27.
- [4] M. S. Floater. "Parametrization and smooth approximation of surface triangulations". In: *Computer aided geometric design* 14.3 (1997), pp. 231–250.
- [5] K. Hormann and G. Greiner. *MIPS: An efficient global parametrization method*. Tech. rep. DTIC Document, 2000.
- [6] K. Hormann, G. Greiner, and S. Campagna. "Hierarchical parametrization of triangulated surfaces". In: *Vision, Modeling, and Visualization*. Vol. 1999. 1999, p. 219.
- [7] A. Iserles. *A first course in the numerical analysis of differential equations*. 44. Cambridge University Press, 1996.



- [8] B. Lévy, S. Petitjean, N. Ray, and J. Maillot. “Least squares conformal maps for automatic texture atlas generation”. In: *ACM Transactions on Graphics (TOG)*. Vol. 21. 3. ACM. 2002, pp. 362–371.
- [9] W. T. Tutte. “How to draw a graph”. In: *Proc Lond Math Soc* 13 (1963), pp. 743–767.
- [10] E. Wachspress. *A Rational Finite Element Basis*. Academic Press rapid manuscript reproductions. Academic Press, 1975.